

Wazuh Homelab Lab 1 – High Severity Vulnerability Remediation on Kali Endpoint

Executive Summary

Built a two-node Wazuh SIEM homelab consisting of a dedicated Wazuh manager VM and a Kali Linux endpoint VM with the Wazuh agent installed. Detected a high-severity vulnerability on the Kali endpoint, investigated the affected package, applied a risk-based remediation strategy, and validated successful resolution in the Wazuh dashboard.

Lab Environment

SIEM Platform: Wazuh

Manager Node: Dedicated Wazuh VM

Endpoint Node: Kali Linux VM

Networking: VirtualBox NAT Network

Endpoint Monitoring: Wazuh Agent connected and active

Objective

Use Wazuh to identify, investigate, and remediate a real vulnerability on a monitored Linux endpoint while practicing SOC analyst workflow.

Detection

The Wazuh dashboard reported a high-severity vulnerability with the following details:

Agent Name: kali

Package Name: weasyprint

Installed Version: 67.0

Severity: High
CVE: CVE-2025-68616

Investigation Process

Reviewed the affected package and determined:

- The software was not intentionally installed for current cybersecurity learning objectives.
 - The package was likely installed as a dependency of another tool.
 - The Kali repository did not provide a newer patched version at the time of review.
 - Retaining unnecessary vulnerable software would increase attack surface.
-

Risk Assessment

Applied the security principle of minimizing attack surface area by removing unnecessary vulnerable software.

Decision: Remove the unnecessary vulnerable package rather than keep unused software installed.

Remediation Actions

Executed the following commands on the Kali endpoint:

```
sudo apt remove weasyprint -y  
sudo apt autoremove -y  
sudo systemctl restart wazuh-agent
```

Validation

After remediation and agent refresh, the Wazuh dashboard updated to:

High Severity Findings: 0

This confirmed the vulnerable package had been successfully removed and the risk was mitigated.

Post-Remediation Hardening

After vulnerability remediation was completed, default Wazuh dashboard administrator credentials were rotated using the official server-side password management tool.

This improved the security posture of the homelab by removing default credentials and validating secure administrative access.

Actions Performed

- Accessed the Wazuh server through SSH
- Identified the official Wazuh password management utility
- Changed the reserved `admin` account password to a strong compliant password
- Tested successful login using the updated credentials

Security Value

- Reduced risk associated with default credentials
 - Improved identity and access management practices
 - Reinforced secure administrative operations within the lab environment
-

Skills Demonstrated

- SIEM Monitoring with Wazuh
 - Vulnerability Detection and Triage
 - Linux Package Management
 - Risk-Based Decision Making
 - Endpoint Security Operations
 - Security Validation and Verification
-

Key Lessons Learned

- Not every vulnerability requires patching first.
- If software is unnecessary, removal can be the safest remediation path.

- Wazuh can be used effectively to monitor endpoint risk posture in a homelab environment.
 - Proper documentation converts labs into portfolio evidence.
-

Portfolio Value Statement

Built and operated a Wazuh SIEM homelab, remediated a high-severity Linux endpoint vulnerability, and validated successful risk reduction through dashboard monitoring.